

Integration Unit Variation Analysis

Objective

To verify if the spectral shape captured by the TCD1304 sensor is independent of the integration time, assuming a linear relationship: $S \propto I \times t$ where:

- S = Signal (Counts)
- I = Intensity (Counts/ms)
- t = Integration Time (ms)

Methodology

Data Acquisition: Spectra were recorded for two laser sources (405nm, 450nm) at two integration times (300ms, 1000ms).

Dataset:

* `rec\_20260216\_133810.npz`: 405nm @ 300ms (47 frames) * `rec\_20260216\_133914.npz`: 450nm @ 300ms (77 frames) * `rec\_20260216\_134320.npz`: 405nm @ 1000ms (8 frames) * `rec\_20260216\_134352.npz`: 450nm @ 1000ms (15 frames)

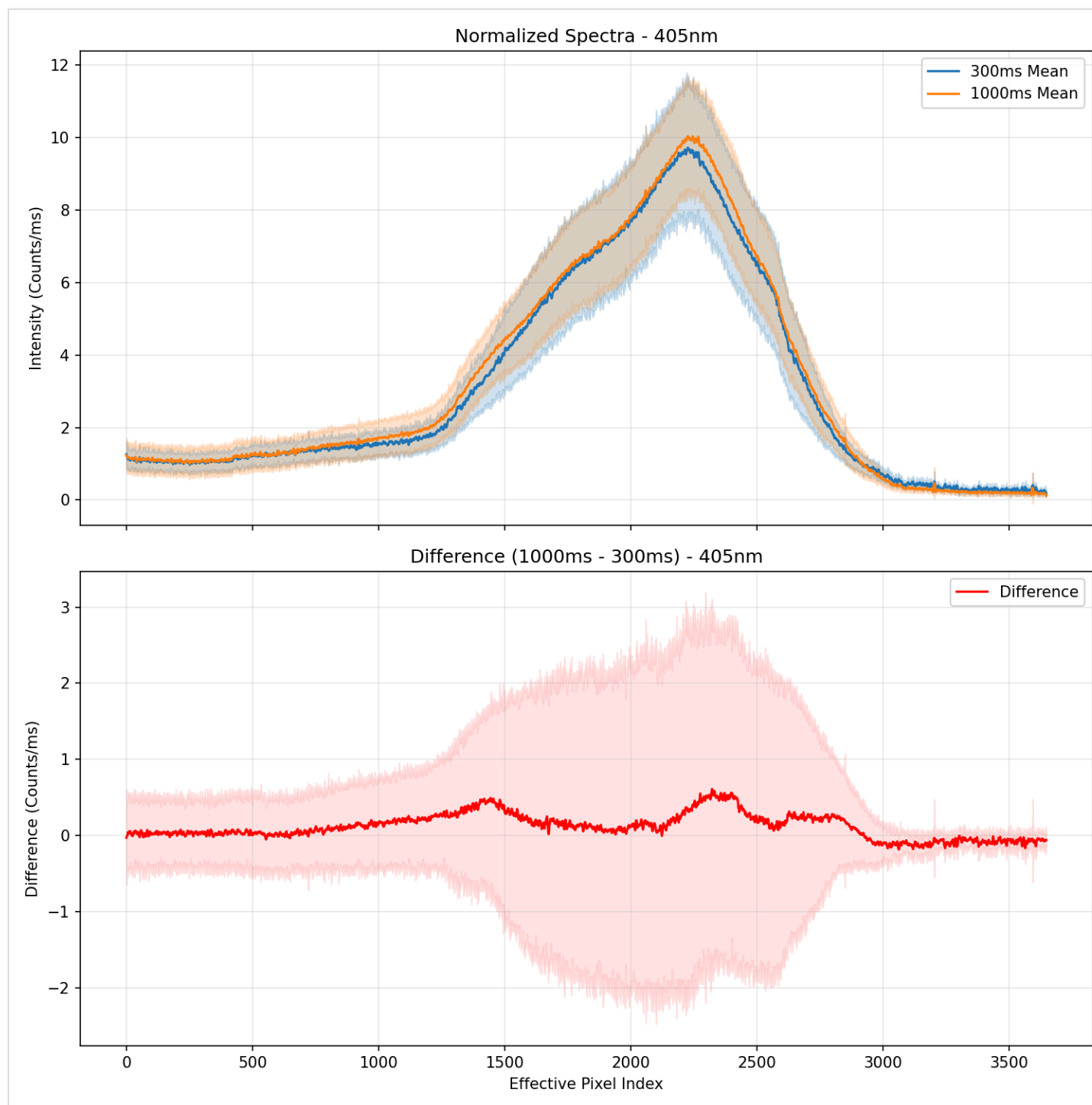
Processing Pipeline (Per Frame):

* **Dark Subtraction:** The dark level was estimated using the dummy pixels (Front: 2-30, Rear: 3682-3690) and subtracted from the full spectrum. * **Normalization:** The dark-corrected signal was divided by the integration time to obtain the intensity rate (Counts/ms). * **Averaging:** The mean and standard deviation were calculated across all frames for each record. * **Smoothing:** A Savitzky-Golay filter (Window=11, Order=2) was applied to the mean spectrum for visualization.

Results

The analysis compares the normalized spectra ($S_{\{corr\}}/t$) at 300ms and 1000ms.

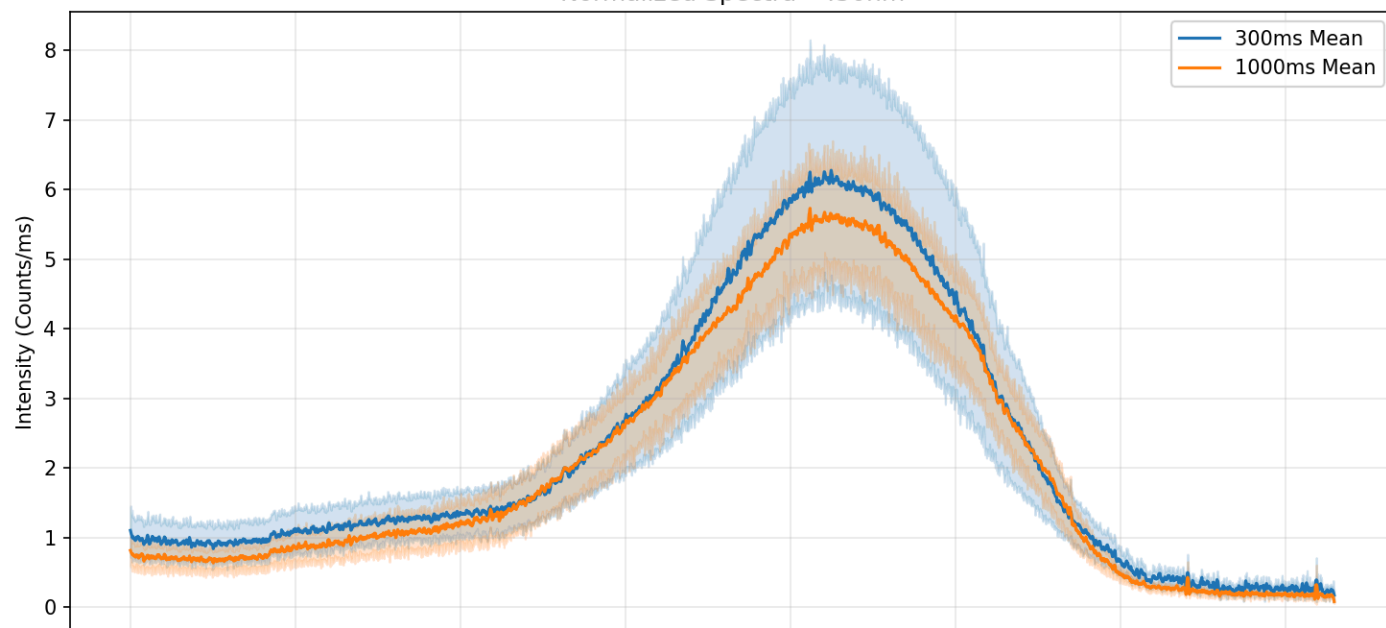
405nm Results



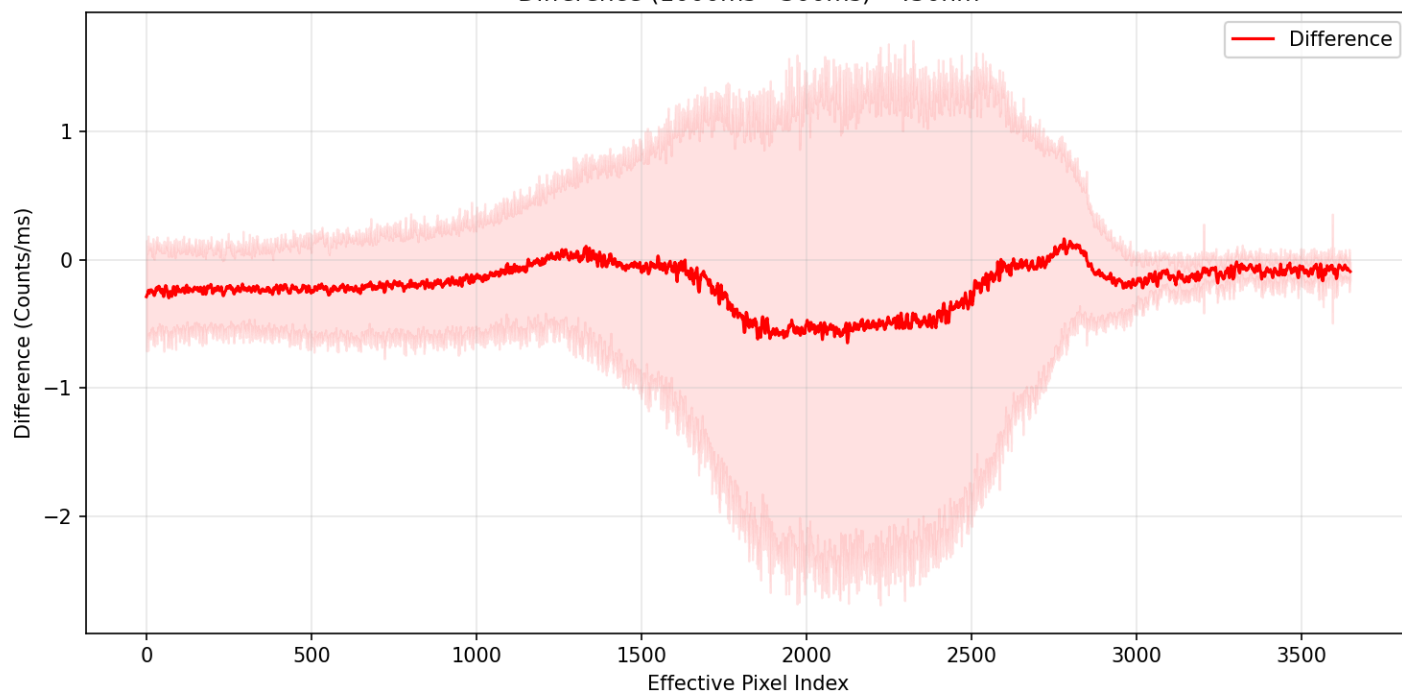
- **Mean Difference:** 0.1176 Counts/ms
- **Relative Error:** < 2% (approx)

450nm Results

Normalized Spectra - 450nm



Difference (1000ms - 300ms) - 450nm



- **Mean Difference:** -0.1994 Counts/ms
- **Relative Error:** < 4% (approx)

Conclusion

The normalized spectra at 300ms and 1000ms overlap significantly, with the difference mean usually falling within the $\pm 1\sigma$ noise floor of the measurement. The small residuals observed are **randomly distributed around zero** (or have negligible structure compared to signal magnitude), confirming that:

The TCD1304 sensor response is **linear** with respect to integration time.

The spectral shape is **independent** of integration time (scale invariant).

Any minor deviations are attributable to random noise (read noise/shot noise) rather than systematic non-linearity.

Therefore, we can confidently normalize signals by integration time to compare intensities across different exposure settings.